

Understanding NAND Gates with CD4011

Understand the working of NAND gates and their application with the help of suitable examples

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CD4011 is the most commonly used complementary metal-oxide semiconductor (CMOS) chip. It comes in a 14-pin dual-inline package (DIP). The small notch on its one side identifies pin 1. There are four independent NAND gates (N1, N2, N3 and N4) in a single chip. Each gate has two inputs and one output. Working voltage range of the IC is 5V to 16V. It can deliver about 10 mA at 12V but this can reduce as the power supply voltage reduces.

Out of the 14 pins of the IC, pins 7

and 14 are connected to the battery or DC power supply. The negative terminal of the battery is connected to pin 7 and the positive terminal to pin 14. In the first gate N1, pins 1 and 2 are for inputs and pin 3 for output (refer Fig. 1). As you can see in the truth table, it will provide an output equal to '1' when any of its inputs is low. In other words, you will receive a high output when any of the input signals goes low. If both the inputs are high, you will receive a low output.

Working of NAND gates

To understand the working, consider the circuit of gates shown in Fig. 2 as an example.

If pins 1 and 2 of N1 is logic high, you will get logic 0 at output pin 3 (refer the truth table). Therefore you will receive logic '1' at output pin 4 because logic '0' is passed to gate N2 from pin 3. So whatever be the input from pin 6, you will receive high at pin 4. This means that you will receive 12V at pin 4. As a result, capacitor starts charging and you will receive logic '1' on pins 8 and 9 and logic '0' at pin 10, which is connected to pin 5. So logic '0' on pin 5 is no longer needed as logic '0' on pin 6 is maintained by gates N2 and N3.

As the capacitor charges further, volt-

age on the resistor drops. When it drops to half the power supply voltage, gate N3 detects a low on its input. As a result, logic '1' appears on its pin 10. As logic '1' is now on pin 5 (since no logic is present on pins 1 and 2), and also on pin 6, you will receive logic 0 at pin 4. As a result, the capacitor starts discharging on pins 8 and 9.

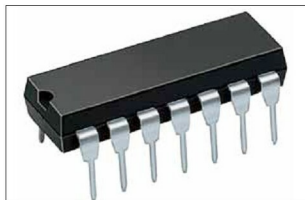
From above, you can see that the circuit works for a certain period of time and then goes off. Time period of operation can be determined by:

$$T=0.7 \times RC$$

Applications

Logic gates have many applications such as in computer circuits:

Door-open indicator. The circuit in Fig. 3 shows how NAND gates are used to activate a light bulb if any of the doors is left open. A door which is properly shut will open a normally-closed pushbutton switch and apply



IC CD4011

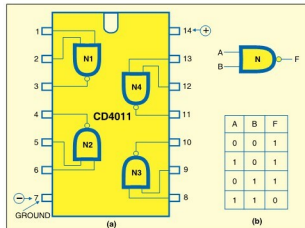


Fig. 1: (a) The internal circuit of CD4011 and (b) truth table for a NAND gate

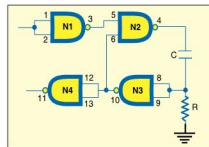


Fig. 2: The circuit of NAND gates

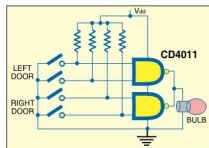


Fig. 3: Circuit for door-open indicator using NAND gates